



ERP

Project Documentation

An AI-powered Enterprise Resource Planning platform for centralized business management, analytics and intelligent decision-making

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01. PROJECT AT A GLANCE

Enterprise Resource Planning is an AI-powered business management system designed to help organizations manage sales, inventory, finance, customers, employees and business reports through one centralized platform.

The system integrates with **Zoho Books** as the primary source of business data and uses **Google Gemini AI** to generate business insights, forecasts, recommendations, executive summaries and AI-assisted analysis.

Core Modules

Dashboard

Sales Management

Inventory Management

Finance Management

Customer Management

Supplier Management

Employee Management

Reports

Analytics

AI Business Insights

Zoho Integrations

Notifications

Global Search

Settings

User Management

Role-Based Access

AI Business Assistant

The platform is designed to reduce manual business analysis, centralize operational information and support faster data-driven decision-making.

02. PROBLEM, USERS & PRODUCT VALUE

Problem

Many businesses manage sales, inventory, finance, customers and reports across different systems.

This creates fragmented data, repetitive manual work, delayed reporting and difficulty monitoring business performance from one place.

The ERP system addresses this by connecting business data through a centralized platform and using AI to provide real-time insights, forecasts and intelligent recommendations.

Users

Administrator

Manages the overall business system, users, roles, integrations and organization-level settings.

Manager

Monitors business performance, dashboards, reports, analytics and AI-generated recommendations.

Finance Team

Reviews financial data, expenses, payments, invoices, tax information and financial reports.

Sales Team

Manages sales, customers, orders, invoices, quotations and sales performance.

Inventory Team

Tracks products, stock levels, warehouses, inventory movement and low-stock alerts.

HR / Employee Management

Manages employee information, departments, attendance, leave, payroll and performance-related information.

Product Value

- Centralized business management
 - Real-time business dashboards
 - Sales and inventory monitoring
 - Financial analysis
 - Employee management
 - Automated reporting
 - AI-powered business insights
 - Predictive analytics
 - Zoho integration
 - Role-based access
-

03. CUSTOMER PERSONA

Primary Persona: Business Owner / Operations Manager

Goal: Monitor and manage important business operations from one centralized platform while using real-time data and AI insights to make better decisions.

Needs

- Monitor sales performance
- Track inventory and stock levels
- Review financial performance
- Monitor employee activity
- Access business reports
- Understand business trends
- Identify operational risks
- Receive AI-powered recommendations
- View important KPIs from one dashboard

Pain Points

- Business data spread across multiple systems
- Manual data analysis
- Delayed reporting
- Difficulty monitoring departments together
- Limited real-time business insights
- Time-consuming decision-making
- Difficulty forecasting business performance

ERP Solution

The ERP platform centralizes business information through a unified dashboard and connects external business data through Zoho services.

04. ARCHITECTURE & TECHNOLOGY

The ERP application uses a modern frontend and backend architecture with external business-data and AI integrations.

Frontend

- React.js
- TypeScript
- Tailwind CSS
- React Router
- Framer Motion
- Recharts / Chart.js
- React Hook Form

Backend

- FastAPI
- Python
- Uvicorn
- Pydantic
- REST APIs
- JWT Authentication
- CORS

Database

PostgreSQL / SQLAlchemy where local persistence is required

External Services

- Zoho Books API
- Zoho CRM API
- Zoho Inventory API
- Google Gemini API

Deployment

- Docker
- Vercel
- Render / Railway

Architecture

User → React / TypeScript → REST API → FastAPI → Zoho Books / Database → Gemini AI → Business Insights

The documented backend structure includes separate API modules for authentication, dashboard, sales, inventory, finance, customers, reports, notifications, business insights and forecasting, along with dedicated Zoho, Gemini and authentication services.

05. CORE ERP WORKFLOW

The main ERP workflow connects authentication, business modules, external data and AI analysis.

**Authentication → Dashboard → Sales / Inventory / Finance / Customers / Employees
→ Reports & Analytics → AI Insights → Business Decisions**

Business Data Workflow

**Zoho Books → ERP Backend → Business Data → Dashboard → Analytics → Gemini
AI → Insights & Recommendations**

Operational Workflow

Users authenticate into the platform and access a centralized dashboard.

From the dashboard, users can navigate between sales, inventory, finance, employee, customer, supplier and reporting modules.

Business data can be synchronized through the Zoho integration layer, while AI analysis provides additional insights and recommendations.

The source project specifically describes Zoho Books as the primary business-data source and Gemini as the AI layer for insights, forecasts and recommendations.

06. FEATURE MAP

Dashboard

Provides an overview of important business KPIs including revenue, sales, orders, inventory value, pending payments, employee performance and AI business score.

Sales Management

Manage orders, quotations, invoices, payments, returns, customers, sales analytics, sales forecasting and sales reports.

Inventory Management

Manage products, categories, warehouses, stock levels, inventory valuation, stock movement, purchase orders, goods receipt, goods dispatch, low-stock alerts and inventory reports.

Finance

Monitor revenue, expenses, profit, cash flow, invoices, payments, taxes, accounts payable, accounts receivable, budgeting and financial reports.

Customer Management

Manage customers, customer profiles, purchase history, segmentation, analytics and customer activity.

Supplier Management

Manage suppliers, contact details, supply performance, payment status, purchase history and supplier ratings.

Employee Management

Manage employee directories, departments, attendance, leave, performance and employee-related information.

Reports

Generate sales, inventory, employee, finance, customer, supplier and AI reports with PDF, Excel and CSV export capabilities.

Analytics

Provide interactive charts, KPI cards, filters and business-performance analysis.

AI Business Insights

Gemini-powered insights including business health score, growth forecasts, revenue prediction, inventory forecasting, demand forecasting, customer churn prediction, expense analysis, risk analysis and recommendations.

Zoho Integration

Connect and synchronize Zoho CRM, Zoho Books and Zoho Inventory through OAuth.

Global Search

Search across products, invoices, customers, employees, reports, CRM records, inventory and AI-related information.

Settings

Manage company profile, users, roles, permissions, security, API keys, Zoho configuration, Gemini configuration, themes, languages, currency, timezone, backups and audit logs.

07. BUILD: DESIGN MODE

Objective

Create a modern, responsive AI-powered ERP platform with a premium enterprise SaaS experience.

The Design Mode direction focused on a professional ERP interface combining dashboards, business analytics, AI insights, Zoho integrations and multiple business modules.

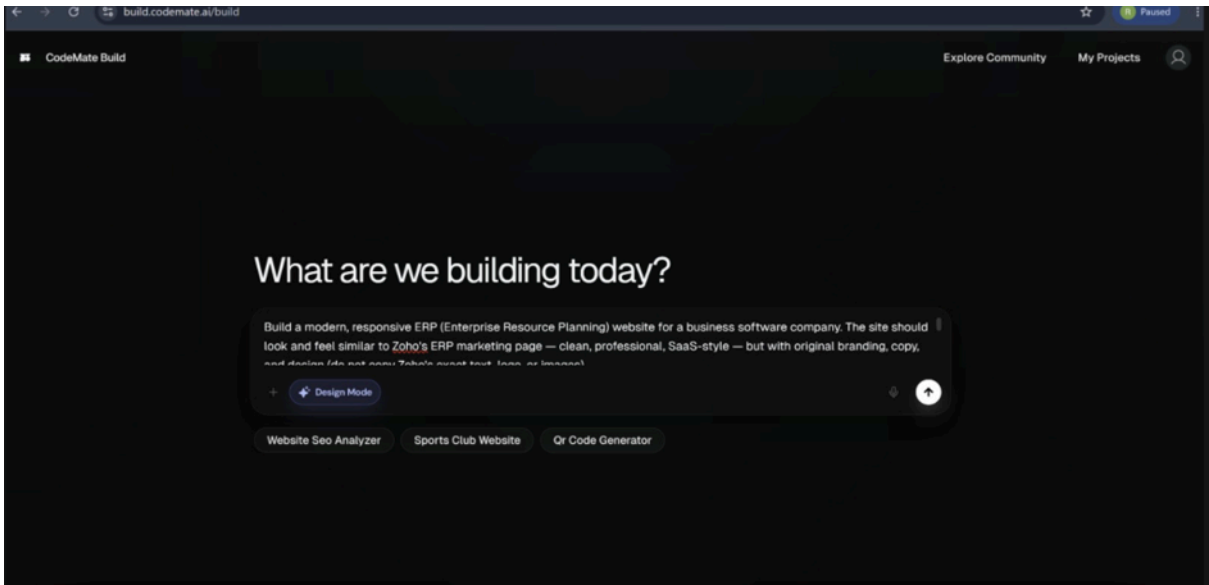
Design Direction

- Premium SaaS interface
- Enterprise ERP visual language
- Dark interface
- Glassmorphism
- Purple and blue accents
- Green success indicators
- Modern dashboards
- Interactive charts
- Responsive layouts
- Professional typography
- Smooth animations
- AI-focused product experience

Design Experience

- Build Design Mode covered the major ERP experience including:
- Landing Page
- Login
- Dashboard
- Sales
- Inventory
- Finance
- Employees
- Customers
- Suppliers
- Reports
- Analytics
- AI Insights
- Integrations
- Settings
- User Profile

The original Design Mode specification also included Zoho integration, ERP module cards, AI dashboards, business insights, pricing, testimonials, FAQ and other landing-page sections.



08. BUILD: PROTOTYPE MODE

Objective

Convert the approved ERP design into a functional frontend application using the defined technology stack and realistic mock business data before backend integration.

Implementation

- Build was used to create:
- Landing Page
- Login
- Dashboard
- Sales Management
- Inventory Management
- Finance Dashboard
- Customer Management
- Supplier Management
- Employee Management
- Reports
- Analytics
- AI Insights
- Integrations
- Settings
- User Profile
- Search
- Filters
- Responsive layouts

Dashboard

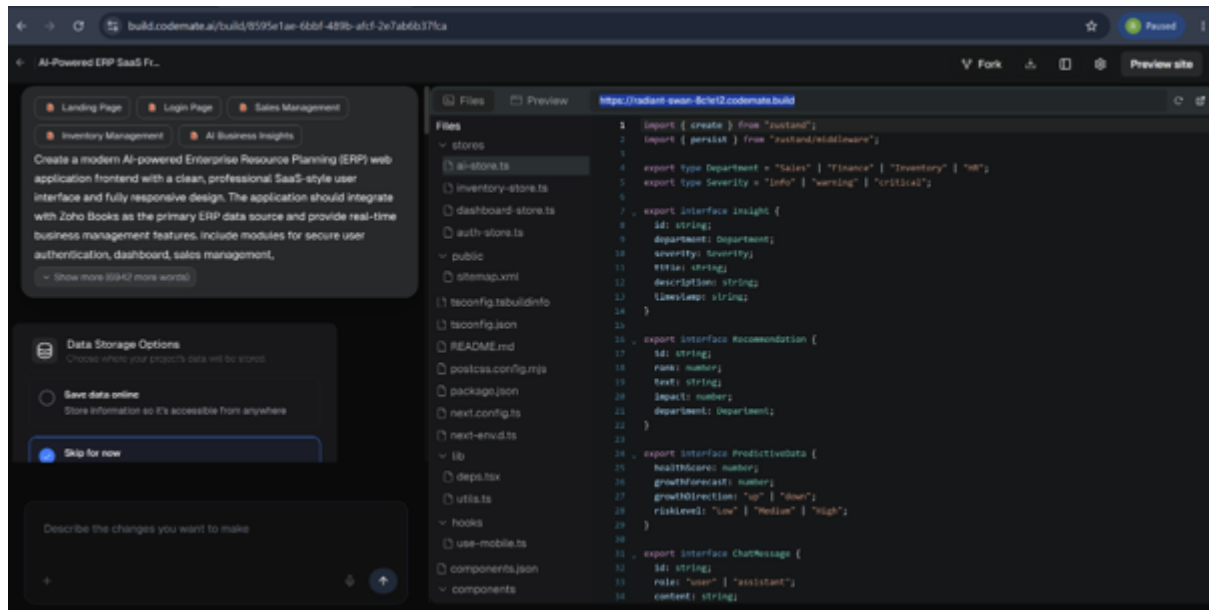
The dashboard includes KPI cards, revenue and sales charts, profit analysis, inventory status, recent activities, orders, notifications and quick actions.

Reusable Components

- Sidebar
- Navbar
- Dashboard Cards
- Charts
- Data Tables
- KPI Cards
- Status Badges
- Forms
- Filters
- Search

- Pagination
- Modals
- Tabs
- Dropdowns
- Notifications
- AI Panels

The Build prompt defined a complete enterprise ERP experience with responsive layouts, reusable components, authentication, business modules, AI functionality and external integrations.



09. BUILD PROMPT HIGHLIGHTS

The Build prompts focused on creating a complete AI-powered ERP application rather than a generic dashboard.

Main Prompt Focus

- AI-powered ERP
- Enterprise SaaS interface
- Sales management
- Inventory management
- Finance
- CRM
- HR
- Analytics
- AI business insights
- Zoho integrations
- Responsive UI
- Role-based access

Prompts Used

Prompt 1 : ERP Design Foundation

Build a modern, responsive AI-powered ERP website for a business software company. Create a clean, premium SaaS experience with a professional marketing website and a complete ERP dashboard after login. Use React, TypeScript, Tailwind CSS, responsive layouts, modern charts and reusable components. The product should centralize sales, inventory, finance, employees, customers, reports and AI-powered business insights into one enterprise platform.

Prompt 2 : ERP Modules & Dashboard

Create a complete enterprise ERP dashboard with a collapsible sidebar and modules for Dashboard, Sales, Inventory, Finance, Employees, Customers, Suppliers, Reports, Analytics, AI Insights, Integrations and Settings. Include KPI cards for revenue, sales, orders, inventory value, pending payments and employee performance, along with revenue trends, sales analytics, profit analysis, inventory status, recent activities, notifications and quick actions. Keep the interface responsive, professional and suitable for a production ERP platform.

Prompt 3 : AI & Zoho Integration

Extend the ERP application with a dedicated AI Business Insights module powered by Google Gemini and a Zoho Integration Center supporting Zoho CRM, Zoho Books and Zoho Inventory. Include AI-powered business health scores, growth forecasts, revenue predictions, inventory forecasting, customer churn analysis, expense analysis, risk indicators, recommendations, natural-language queries and executive summaries. Add OAuth-based Zoho connection management, synchronization status, last sync time, sync progress, errors, manual sync, automatic sync and secure token handling.

Prompt 4 : Complete Functional ERP

Build the finalized ERP design into a production-ready application while preserving the approved visual system. Implement reusable components, authentication, role-based access, dashboards, sales, inventory, finance, CRM, HR, customer and supplier management, reports, analytics, AI insights, Zoho integrations, global search, notifications and settings. Ensure the application is responsive across desktop, tablet and mobile devices, includes proper loading and error states, and maintains clean modular architecture throughout.

10. CORA: PLANNING & IMPLEMENTATION

CORA was used as the main development and implementation layer for transforming the ERP frontend into a backend-connected application.

CORA supported backend development, API creation, code improvements, debugging, integration work and implementation of AI-powered business-management functionality.

CORA was used for:

- Backend setup
- FastAPI architecture
- API development
- Authentication
- Zoho Books integration
- Gemini AI integration
- Dashboard APIs
- Sales APIs
- Inventory APIs
- Finance APIs
- Customer APIs
- Reports
- Business Insights
- Forecasting
- Notifications
- Database connectivity
- Testing and debugging

Workflow

Build Frontend → CORA Planning → FastAPI Backend → API Integration → Zoho / Gemini → Testing → Final Application

Prompts Used

Prompt 1 : Initial Backend Setup

Analyze the existing React + TypeScript ERP frontend project and generate a production-ready FastAPI backend for an AI-powered Enterprise Resource Planning system. Integrate Zoho Books API as the primary ERP data source and Google Gemini AI API for business insights. Do not modify the existing frontend. Use Python 3.11+, FastAPI, Uvicorn, Pydantic, JWT Authentication, Passlib, Python-dotenv, HTTPX or Requests, CORS, Zoho Books API, Gemini API and Docker. Create a modular backend structure with API modules for authentication, dashboard, sales, inventory, finance, customers, reports, notifications, business insights and forecasting.

Prompt 2 : Zoho Integration

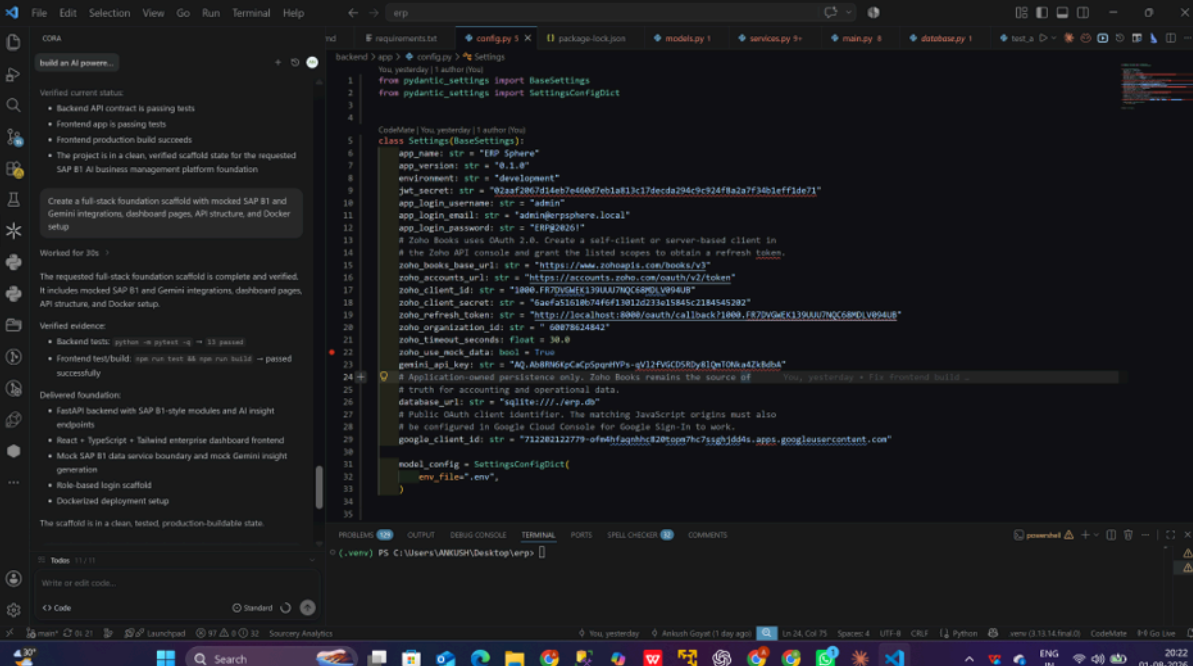
Implement the Zoho integration layer for the ERP backend. Connect Zoho Books as the primary source of business data and provide support for Zoho CRM and Zoho Inventory where required. Implement OAuth-based authentication, secure token handling, connection status, synchronization status, last sync time, manual synchronization, automatic synchronization, error handling and module-level data retrieval. Structure the integration as a dedicated service so that the dashboard, sales, inventory, finance and reporting APIs can consume synchronized business data without duplicating the integration logic.

Prompt 3 : Gemini AI Business Insights

Implement the AI business-insights layer using the Google Gemini API. Create a dedicated Gemini service that can analyze ERP business data and generate business health scores, growth forecasts, revenue predictions, inventory forecasts, demand forecasts, expense analysis, risk indicators, recommendations, executive summaries and automated business reports. The AI assistant should also support natural-language business questions and return useful explanations based on the available ERP data.

Prompt 4 : API & Application Integration

Connect the ERP frontend with the newly implemented FastAPI backend while preserving the existing UI. Integrate the dashboard, sales, inventory, finance, customers, reports, AI insights and authentication APIs. Verify API requests and responses, handle loading and error states, validate authenticated access, ensure Zoho data is retrieved correctly, and verify that Gemini-generated insights are displayed correctly inside the application. Test the complete frontend-to-backend-to-external-service workflow after integration.



```
1 from pydantic_settings import BaseSettings
2 from pydantic_settings import SettingsConfigDict
3
4
5 class Settings(BaseSettings):
6     app_name: str = "ERP Sphere"
7     app_version: str = "0.1.0"
8     environment: str = "Development"
9     jwt_secret: str = "82af20c7d14eb7e4d4b7eb1e813c17decda294c9c24f8a2a7f34b1eff1de71"
10     app_login_username: str = "admin"
11     app_login_email: str = "admin@erpsphere.local"
12     app_login_password: str = "ERP@2026!"
13     # Zoho Books uses OAuth 2.0. Create a self-client or server-based client in
14     # the Zoho API console and grant the listed scopes to obtain a refresh token.
15     zoho_books_base_url: str = "https://www.zohoapis.com/books/v3"
16     zoho_accounts_url: str = "https://accounts.zoho.com/oauth/v2/token"
17     zoho_client_id: str = "1800.F87DGMH139XUJ7NQC68MDV094UP"
18     zoho_client_secret: str = "6ae451618074f6f13012023be15845c2184545282"
19     zoho_refresh_token: str = "http://localhost:8080/oauth/callback?1800.F87DGMH139XUJ7NQC68MDV094UP"
20     zoho_organization_id: str = "60878624842"
21     zoho_timeout_seconds: float = 30.0
22     zoho_use_mock_data: bool = True
23     # Application-owned persistence only. Zoho Books remains the source of
24     # truth for accounting and operational data.
25     database_url: str = "sqlite:///erp.db"
26     # Public OAuth client identifier. The matching JavaScript origins must also
27     # be configured in Google Cloud Console for Google Sign-In to work.
28     google_client_id: str = "712282122779-ofadhfaqbhg820tqm7hc7tqghjdd46.apps.googleusercontent.com"
29
30     model_config = SettingsConfigDict(
31         env_file=".env",
32     )
33
34
35
```

11. CORA TASK ARCHIVE

Instead of listing every individual CORA task, the implementation can be grouped into the following major areas.

Backend

- FastAPI backend
- Uvicorn
- Pydantic
- REST APIs
- API routing
- Services
- Middleware
- Configuration
- Security
- Docker

API Modules

- Authentication
- Dashboard
- Sales
- Inventory
- Finance
- Customers
- Reports
- Notifications
- Business Insights
- Forecasting

Integrations

- Zoho Books
- Zoho CRM
- Zoho Inventory
- Google Gemini AI

AI

- Business Health Score
- Growth Forecast
- Revenue Prediction
- Inventory Forecasting
- Demand Forecasting
- Customer Churn Prediction
- Expense Analysis
- Risk Analysis
- AI Recommendations
- Executive Summaries
- Natural Language Queries

Business Modules

- Sales
- Inventory
- Finance
- CRM
- Customers
- Suppliers
- Employees
- Reports
- Analytics

Security

- JWT authentication
- Role-based access
- Protected routes
- Password security
- Secure environment variables
- API security

Testing & Debugging

- API request validation
- Response validation
- Integration testing
- Authentication testing
- Zoho data verification
- Gemini response verification
- Dashboard validation

The project documentation confirms that CORA was used for code generation, debugging, API endpoints, architecture improvements and integration of FastAPI, React, Zoho Books and Gemini AI.

12. IMPLEMENTATION MILESTONES

01. Design

Created the initial ERP product experience using Build Design Mode.

02. Frontend

Converted the approved design into a responsive React and TypeScript application with dashboards, business modules, analytics and AI interfaces.

03. Backend

Created the FastAPI backend architecture with modular API and service layers.

04. Authentication

Implemented secure authentication, JWT-based access and role-based access requirements.

05. Zoho Integration

Integrated Zoho Books as the primary business-data source with support for Zoho CRM and Zoho Inventory.

06. Business Modules

Implemented the main ERP areas including sales, inventory, finance, customers, suppliers, employees and reports.

07. AI

Integrated Google Gemini for business insights, forecasts, recommendations and AI-assisted analysis.

08. Analytics & Reporting

Added interactive dashboards, business analytics, reporting and data visualization.

09. Verification

Tested API communication, Zoho data retrieval, dashboard statistics, reports, AI insights, authentication and frontend-backend integration.

13. BACKEND, DATA & INTEGRATIONS

The ERP backend is built around FastAPI and provides API and service layers for business operations, external integrations and AI functionality.

Core Data Areas

- Users
- Sales
- Orders
- Invoices
- Products
- Inventory
- Customers
- Suppliers
- Employees
- Finance
- Expenses
- Payments
- Reports
- Analytics
- Business Insights

Backend Flow

Frontend → REST API → FastAPI → Service Layer → Zoho / Database → Response → Dashboard

Zoho Flow

Zoho Books / CRM / Inventory → OAuth → Integration Service → ERP APIs → Dashboard

AI Flow

Business Data → Gemini Service → AI Analysis → Forecast / Recommendation → Dashboard

The documented backend structure separates API modules such as dashboard, sales, inventory, finance, customers, reports and business insights from services for Zoho Books, Gemini and authentication.

Deployment

The project documentation specifies Docker-based deployment with cloud deployment support such as Render.

14. AUTHENTICATION & USER MANAGEMENT

The ERP platform includes authentication and role-based access as part of its core application architecture.

Authentication Flow

Login / Signup → Authentication → JWT → Protected Routes → Role-Based Access → ERP Dashboard

Authentication Features

- User Login
- User Signup
- JWT Authentication
- Refresh Token Support
- Protected Routes
- Role-Based Access
- Secure Session Management
- Password Security

User Roles

- Admin
- Manager
- Finance
- Sales
- Inventory
- HR
- Employee

The detailed ERP specification defines customizable permissions for these roles and includes role-based access as part of the platform security model.

User Management

- Profile
- Role
- Department
- Permissions
- Activity Logs
- Preferences
- Notification Settings
- Security Settings

15. RESPONSIVE UI & POLISH

The ERP interface was designed as a responsive enterprise SaaS application with a premium dashboard experience.

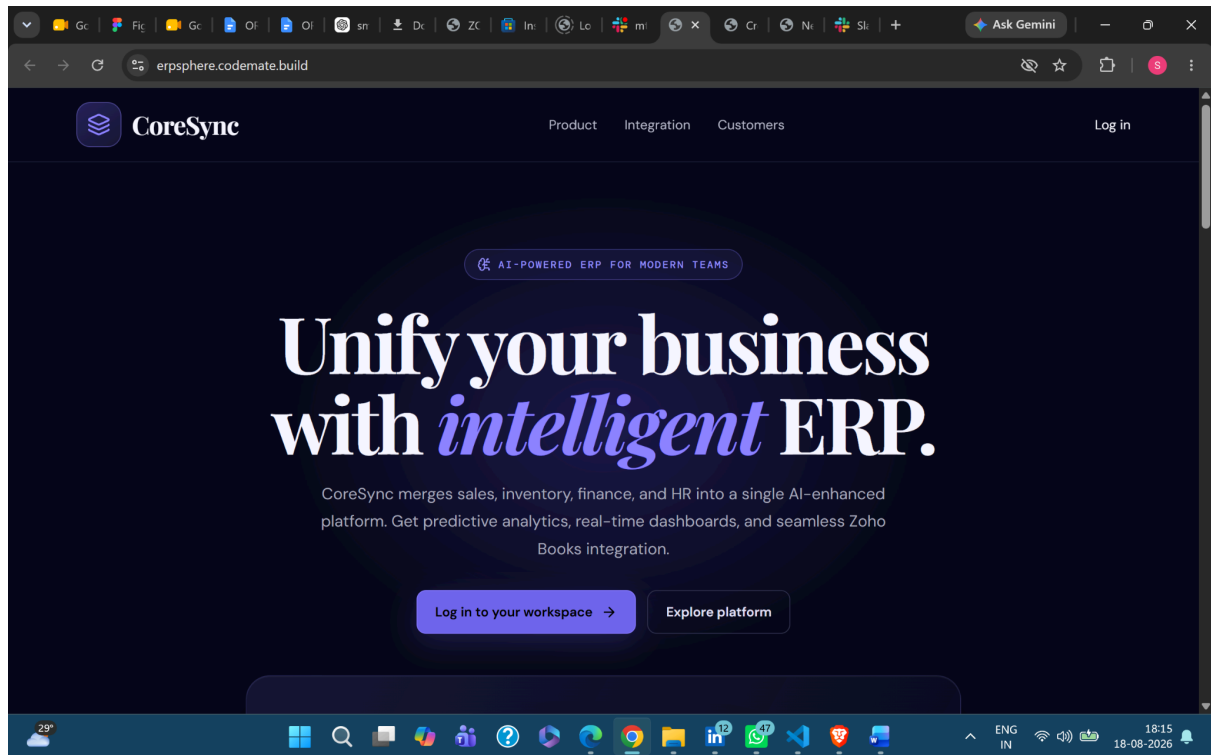
UI Characteristics

- Responsive layouts
- Dark premium interface
- Glassmorphism cards
- Purple and blue accents
- Green success indicators
- Rounded components
- Soft shadows
- Smooth animations
- Interactive charts
- Modern typography
- Reusable UI components
- Loading states
- Empty states
- Error handling
- Accessible interactions

The source design specification explicitly includes responsive behavior across desktop, tablet and mobile along with modern SaaS styling, glassmorphism, rounded cards, shadows and animations.

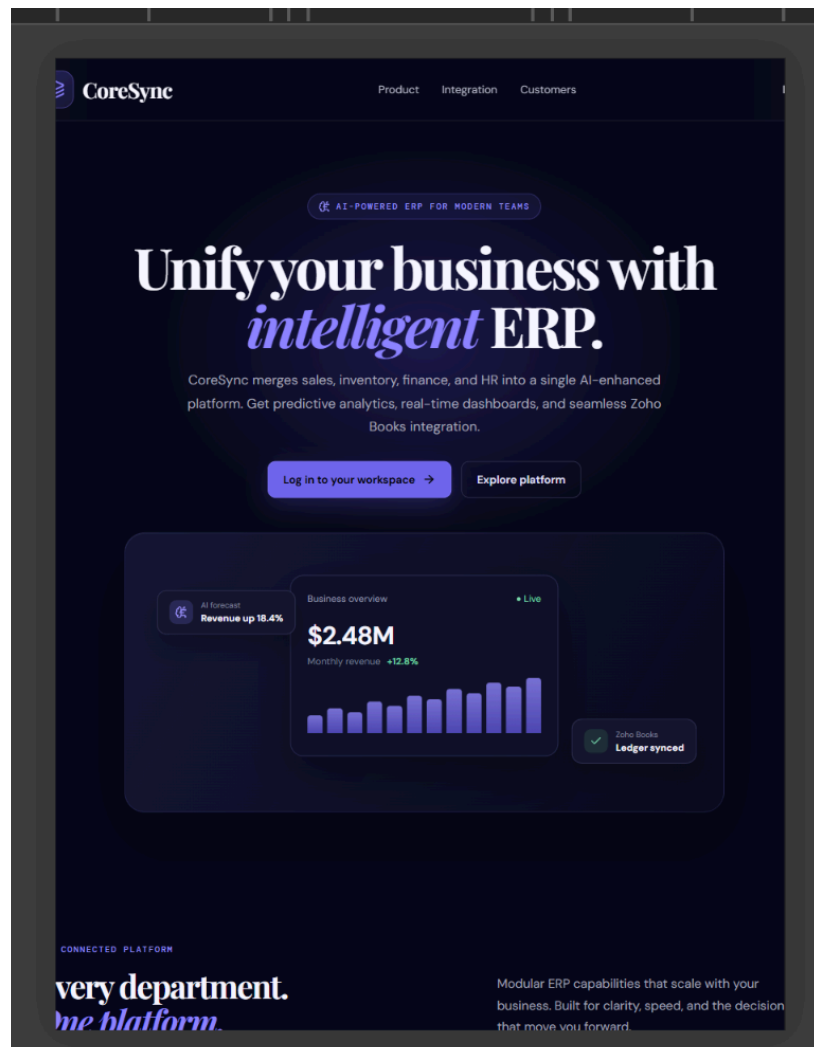
Desktop

Full sidebar navigation, dashboards, charts, data tables and multi-column business modules.



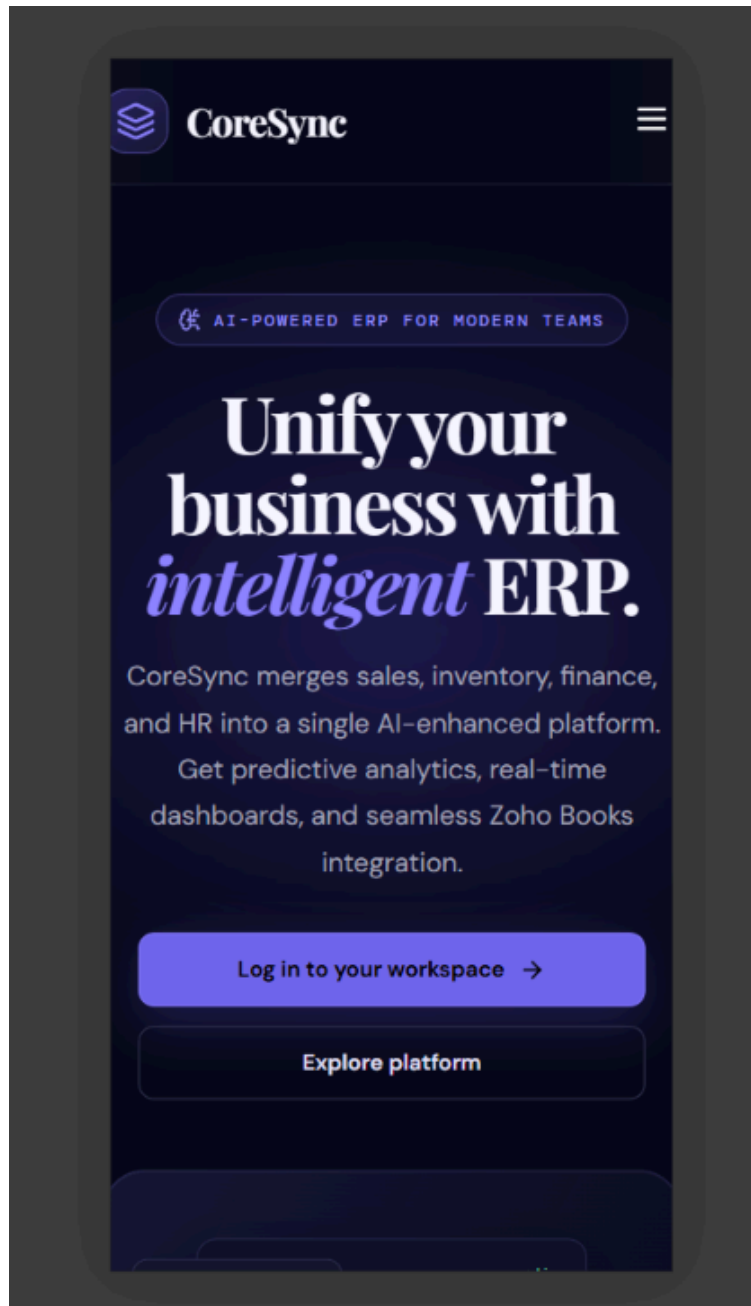
Tablet

Adjusted navigation, content density and responsive dashboard layouts.



Mobile

Collapsed navigation, stacked layouts, responsive cards and touch-friendly interactions.



Final Product Flow

Build Design Mode → Build Code Mode → CORA Development → Zoho Integration → Gemini AI → Final ERP Product

The completed system combines centralized business management, real-time business data, reporting and AI-powered decision support in one platform.
